

SANKHA SUBHRA MAHANTI

Green's Postdoctoral Scholar, IGPP, Scripps Institute of Oceanography, UC San Diego
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EDUCATION

- **Ph.D. in Geosciences** August 2021 - May 2026
Department of Geosciences, University of Arizona Tucson, United States
 - Advisor: Dr. Eric Kiser
- **Integrated BS-MS in Physics** July 2016- July 2021
Department of Physical Sciences, Indian Institute of Science Education and Research (IISER) Kolkata Kolkata, India
 - MS Advisor: Dr. Supriyo Mitra

EXPERIENCE / AFFILIATIONS

- **Green's Postdoctoral Scholar** July 2026 - Present
IGPP, Scripps Institute of Oceanography, UC San Diego, San Diego, United States
- **Graduate Research Associate** May 2023 - May 2026
Department of Geosciences, University of Arizona, Tucson, United States
- **Graduate Research Assistant** January 2022 - December 2022
Department of Geosciences, University of Arizona, Tucson, United States
- **MS Student Researcher** August 2019 - June 2021
Computational Seismology Lab, IISER Kolkata, Kolkata, India
- **Summer Intern** June 2019 - July 2019
Earthquake Research Institute, University of Tokyo, Tokyo, Japan
- **Summer Intern** May 2018 - July 2018
Computational Seismology Lab, IISER Kolkata, Kolkata, India

PUBLICATIONS

Journal Articles

- [1] Bradford, J., **Mahanti, S. S.**, Kiser, E., Beck, S. L., Fernandez, M., Trad, S., Porter, R., Tauber, S., Maharaj, A., Howe, H., Ortiz, G., & Saez, M., (2026) *Transcrustal magma ascent beneath Laguna del Maule, Chile*. *Nature Communications Earth & Environment*, 2026
<https://doi.org/10.1038/s43247-026-03863-8>
- [2] Bradford, J., Beck, S., Kiser, E., Porter, R., **Mahanti, S. S.**, Tauber, S., Fernandez, M., Ortiz, G., Saez, M., et al. (2026) *The anatomy of a modern cordillera style mountain system: Northern Chile and Argentina*. *Journal of Geophysical Research: Solid Earth*, 131, e2026JB034475.
<https://doi.org/10.1029/2026JB034475>
- [3] **Mahanti, S. S.**, Bradford, J., Kiser, E., Beck, S. L., Fernandez, M., Tauber, S., Porter, R., Ortiz, G., & Saez, M., (2026), *Detachment-Involved Deformation in the Santa Barbara System in NW Argentina Revealed by Seismicity and Receiver Functions*. *Geology*, 2026.
<https://doi.org/10.1130/G53942.1>
- [4] **Mahanti, S. S.**, Kiser, E., Beck, S. L., & Hughes, A. N., (2024), *Seismicity and Present Day Crustal Deformation in the Southern Puna Plateau*. *Journal of Geophysical Research: Solid Earth*, 129, e2024JB028918. <https://doi.org/10.1029/2024JB028918>

Articles in Preparation

- [1] **Mahanti, S. S.**, Kiser, E., et al., *High-Frequency Seismic Imaging of the Crust Beneath the Puna Plateau of the Central Andes using Local Earthquake Autocorrelation* (In Prep.)
- [2] **Mahanti, S. S.**, Kiser, E., et al., *Orogen Parallel Variations in Seismicity along the Central Andes and Implications on Crustal Deformation* (In Prep.)

SKILLS

- **Programming & Scripting Languages:** Python, Matlab, Fortran, Shell
- **Research Skills:** Earthquake Detection, Seismic Imaging, Machine Learning (Earthscope certified), High Performance Computing (HPC), Seismic Source Modeling
- **Languages:** English, Bengali, Hindi

CONFERENCE PRESENTATIONS

- [1] Mahanti, S. S., Kiser, E., Bradford, J., Beck, S., *et al.* **Seismic Imaging of Southern Central Andes by Seismic Autocorrelation.** *Poster presentation in SSA Annual Meeting 2025.*
- [2] Mahanti, S. S., Kiser, E., Bradford, J., Beck, S., *et al.* **Orogen Parallel Variations in Seismicity in the Central Andes Recorded by the TANGO Seismic Deployment.** *Poster presentation in AGU Fall Meeting 2024.*
- [3] Mahanti, S. S., Kiser, E., Bradford, J., Beck, S., *et al.* **Preliminary Earthquake Catalog of the Southern Central Andes (23-24°S) Recorded by the TANGO Seismic Deployment.** *Poster presentation in AGU Fall Meeting 2023.*
- [4] Mahanti, S. S., Kiser, E., & Beck, S. L. (2022). **Crustal Seismicity in the Southern Puna Plateau and Relation With Shallow Crustal Structures.** *Oral presentation in AGU Fall Meeting 2022.*
- [5] Mahanti, S. S., Mitra, S., & Miyake, H. (2020). **Teleseismic Source Modelling of Strong-to-major Himalayan Earthquakes.** *Virtual poster presentation in AGU Fall Meeting 2020.*

HONORS AND AWARDS

- **Best Geophysics Talk** | Annual symposium of UAGeosciences 2026
- **Best Geophysics Talk** | Annual symposium of UAGeosciences 2025
- **Invited Speaker at TEDx University of Arizona** 2024
- **Winner of University of Arizona GradSlam** (Best 3-minutes research presentation) 2024
- **Best Geophysics Talk** | Annual symposium of UAGeosciences 2023
- **Best Geophysics Talk** | Annual symposium of UAGeosciences 2022
- **Selected in Sakura Science Exchange Program** | Japan Science and Technology (JST) 2019

GRANTS AND FELLOWSHIPS

- **Green Postdoctoral Fellowship** (IGPP, UC San Diego) 2026
- **Galileo Circle Scholarship** (University of Arizona) 2025
- **Peter Coney Fellowship** (Department of Geosciences, University of Arizona) 2025
- **Sumner, John, and Nancy Scholarship** (Department of Geosciences, University of Arizona) 2024
- **Society of Exploration Geophysicists (SEG) Scholarship** (For academic year 2024-2025) 2024
- **UA GPSC Travel Grant** (For attending AGU Fall Meeting 2023) 2023
- **Society of Exploration Geophysicists (SEG) Scholarship** (For academic year 2022-2023) 2022
- **UA GPSC Travel Grant** (For attending AGU Fall Meeting 2022) 2022
- **Sumner, John, and Nancy Scholarship** (Department of Geosciences, University of Arizona) 2021
- **INSPIRE Scholarship** (Department of Science and Technology, India) 2016-2021

FIELDWORK EXPERIENCE

- **Nodal seismometer deployment for active-source seismic experiment in northern Arizona**2024
Led one team and deployed ~100 nodal seismometers.
- **Nodal seismometer deployment in northern Arizona**2023-2024
Deployed and maintained (monthly swapping) a 2D array of ~46 nodal seismometers for a year.
- **Nodal seismometer deployment in Argentina**2022
Deployed ~150 nodal seismometers on a linear array as a part of the TANGO seismic deployment.

TEACHING EXPERIENCE

- **Graduate Teaching Assistant (Course: Introduction to Geophysics)**Spring 2025
Department of Geosciences, University of Arizona, United States
- **Graduate Teaching Assistant (Course: Introduction to Oceanography)**Spring 2023
Department of Geosciences, University of Arizona, United States
- **Graduate Teaching Assistant (Course: Introduction to Geophysics)**Fall 2021
Department of Geosciences, University of Arizona, United States
- **Teaching Assistant (Course: Thermal Physics)**Spring 2021
Department of Physical Sciences, IISER Kolkata, India
- **Teaching Assistant (Course: Mathematical methods II)**Fall 2020
Department of Physical Sciences, IISER Kolkata, India

VOLUNTEER SERVICE

- **Manuscript reviewer: JGR Solid Earth**2026
- **Student mentor** for the Association for Women Geoscientists (AWG), University of Arizona2025
- **Grant application reviewer** for GPSC, University of Arizona2024